Applying the PAY Rule™ with Monte Carlo Simulations for the Current Market Environment

Next, I simulate these strategies with Monte Carlo simulations for stock and bond returns using current market environment as a starting point. This approach is described in the technical appendix to chapter 3. I believe the results in Exhibit 6.21 are much more realistic than historical data for today's retirees. These simulations reflect the lower bond yields available to retirees today, but a mechanism is included for interest rates to gradually increase over time, on average. Bond returns are calculated from the simulated interest rates and their changes, and stock returns are calculated by adding a simulated equity premium on top of the simulated interest rates. All strategies are simulated with the same asset allocations and portfolio returns for the sake of comparison. Strategies are simulated with annual data, assume withdrawals are made at the start of each year, and use annual rebalancing to restore the targeted asset allocation. The tax implications for different spending strategies are not otherwise considered.

The PAY Rule used to create Exhibit 6.21 is the same as with Exhibit 6.20: The retiree accepts a 10 percent chance that real wealth falls below $15,000 (based on an initial $100,000) by year thirty of retirement. Spending levels in Exhibit 6.21 are generally lower than found with historical data, even though spending patterns are largely similar. Due to the similarities, I will not go into great detail in describing the numbers.

That being said, the starting points merit some discussion. For instance, Bengen's baseline constant inflation-adjusted spending rule supports an initial spending rate of 2.78 percent as a starting point—noticeably lower than the 4.18 percent found with historical data, but it is the implication of retiring in a low interest rate world. For the $100,000 initial portfolio, this supports $2,780 in real spending across the entire distribution of Monte Carlo simulations for as long as wealth remains.

The fixed-percentage strategy allows initial spending to increase to 6.60 percent, with noticeable spending reductions and aggressive wealth depletion over time. Again, the smoothing endowment formula provides a